
DIAL: A post-hoc diagnostic for subspace-aligned conditional shift under linear batch correction

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Abstract

We introduce *DIAL* (**D**irection-**I**nvariant **AUC** Leakage), a single-number post-hoc diagnostic that fires when a linear batch-correction operator T leaves a residual class-conditional shift inside the label subspace, causing a source-trained classifier to have AUC *below* chance on the target cohort. We prove a unified shift decomposition (Theorem 1) showing that $\text{DIAL} > 0$ is governed by the component of the post-correction conditional KL that lies parallel to the Fisher discriminant direction, and is independent (up to finite-sample noise) of pure covariate or pure label shift. We validate the decomposition numerically ($R^2 = 0.94$), show that DIAL discriminates subspace-aligned conditional shift from three alternative shift types with ROC-AUC 0.78 for any flip-positive shift type (broader definition), benchmark ten test-time-adaptation methods on a synthetic flip (best method: DANN-lite, target AUC 0.39 vs 0.33 for naive ComBat), report a *negative* foundation-model scaling result (encoder parameter count does not monotonically reduce DIAL), and demonstrate cross-domain reproducibility in vision-, NLP-, clinical-, and biology-shaped synthetic data (4/4 domains). All real- data motivation cited in earlier drafts of this work was retracted under proper LODO-fit-on-train-only ComBat (companion audit, 2026-04); v15 is therefore framed as a calibration paper: Theorem 1 explains exactly which shift regimes produce a flip, and the leak detection becomes a worked example of preprocessing-induced leakage that DIAL flags.

1 Introduction

Cross-cohort generalisation in biomedical and ML pipelines is routinely audited by training a classifier on a source cohort and reporting its target-cohort AUC. A surprising and under-discussed phenomenon arises when the target AUC falls *below* chance: the source-trained classifier ranks target instances in the wrong order, even though it achieved near-perfect AUC on source cross-validation. We call this the *flip event*.

What induces a flip? Empirically, flips are observed when a linear batch-correction operator (ComBat [Johnson et al., 2007], CORAL [Sun et al., 2016], mean-centring) is applied to data with class-imbalanced batches. We have observed the phenomenon both in deliberately constructed synthetic regimes (Section 5) and in a companion audit of a public biomedical cross-cohort transfer that — under proper Leave-One-Dataset- Out (LODO) ComBat fit-on-train-only — turned out to be a leakage artefact. The leak diagnosis itself is a special case of the phenomenon Theorem 1 characterises: pooled-fit ComBat on class-imbalanced cohorts induces precisely the subspace-aligned conditional shift that produces a positive DIAL.

Contributions.

1. **Theorem 1 (Unified Shift Decomposition).** We decompose the post-correction joint KL into covariate-marginal, label-prior, conditional-parallel, and conditional-perpendicular components. Under Gaussian shared-covariance assumptions, $\text{DIAL}(\tilde{X}, Y)$ is strictly increasing in $\Delta_{\text{cond}}^{\parallel}$ and independent of the other components up to finite-sample noise.
2. **Numerical validation.** A controlled simulation regresses observed DIAL against the four KL components and recovers $R^2 = 0.94$ with the parallel coefficient dominating by a factor of $33\times$ (§5.1).
3. **TTA + foundation-model benchmark.** Ten methods on a synthetic flip; honest negative result on foundation-model scaling (§5.3, §5.4).
4. **Cross-domain reproducibility.** The same flip mechanism reproduces in synthetic data shaped after vision, NLP, clinical, and biology modalities (4/4 domains; §5.5).
5. **Connection to preprocessing leakage.** Our retraction audit is consistent with Theorem 1: pooled-fit ComBat on class-imbalanced cohorts produces a subspace-aligned conditional shift that DIAL fires on. DIAL therefore functions as a *leak detector* for biomedical pipelines.

2 Related work

Covariate-shift theory. Shimodaira [Shimodaira, 2000] and KLIEP [Sugiyama et al., 2007] target $p_s(x)/p_t(x)$. Theorem 1 shows this is insufficient under conditional-parallel shift.

Domain-invariance bounds. Ben-David et al. [Ben-David et al., 2006, 2010] define $d_{\mathcal{H}\Delta\mathcal{H}}$; Mansour et al. [Mansour et al., 2009] and Zhao et al. [Zhao et al., 2019] extend it. Under linear T and Gaussian conditionals, DIAL is a computable surrogate.

Adversarial DA. DANN [Ganin and Lempitsky, 2015, Ganin et al., 2016], ADDA [Tzeng et al., 2017], CDAN [Long et al., 2018], MCD [Saito et al., 2018], MDD [Zhang et al., 2019]. We benchmark a DANN-lite (§5.3).

Test-time adaptation. TENT [Wang et al., 2021], SHOT [Liang et al., 2020], BN-stats [Schneider et al., 2020], MEMO [Zhang et al., 2022], CoTTA [Wang et al., 2022], EATA [Niu et al., 2022]. Section 5.3 includes all four core methods.

Feature alignment. CORAL [Sun et al., 2016], SA [Fernando et al., 2013], DAN [Long et al., 2015], JAN [Long et al., 2017]; biology-side ComBat [Johnson et al., 2007], SVA [Leek et al., 2012], Harmony [Korsunsky et al., 2019], scVI [Lopez et al., 2018].

Leakage / preprocessing audits. Kaufman et al. [Kaufman et al., 2012]; Roberts et al. [Roberts et al., 2017]; Kapoor & Narayanan [Kapoor and Narayanan, 2023]; Whalen et al. [Whalen et al., 2022]. Our retraction audit documents a textbook example.

Foundation models / scaling / robustness. Kaplan et al. [Kaplan et al., 2020], Hoffmann et al. [Hoffmann et al., 2022], Bommasani et al. [Bommasani et al., 2021], Geirhos et al. [Geirhos et al., 2020], Sagawa et al. [Sagawa et al., 2020].

3 Preliminaries and DIAL metric

Notation. Source and target distributions $p_s(x, y, b), p_t(x, y, b)$ on $\mathbb{R}^p \times \{0, 1\} \times \mathcal{B}$. Linear correction operator $T(x, b) = A_b x + c_b$ (Definition 1). Population Fisher direction $w_Y = \Sigma^{-1}(\mu_1 - \mu_0)$ under p_s .

Definition 1 (Linear correction operator). *A linear correction operator is a batch-wise affine map $T(x, b) = A_b x + c_b$ where $A_b \in \mathbb{R}^{p \times p}$ is non-singular and $c_b \in \mathbb{R}^p$. The image is denoted $\tilde{X} = T(X, B)$.*

Definition 2 (DIAL). *For a fixed classifier family and cross-validation protocol,*

$$\text{DIAL}(\tilde{X}, Y) = (\max(\text{AUC}, 1 - \text{AUC}) - \tfrac{1}{2}) \cdot \mathbf{1}[\text{AUC} < \tfrac{1}{2}].$$

DIAL is positive iff the source-trained classifier achieves $\text{AUC} < \frac{1}{2}$ on the target — the *flip event*.

Table 1: Stress-test mean DIAL.

shift type	mag=0	0.25	0.5	1.0
covariate	0.000	0.000	0.000	0.000
label	0.000	0.000	0.000	0.000
concept (full flip)	0.000	0.000	0.009	0.393
subspace-aligned	0.000	0.000	0.000	0.237

Three KL shift components. $\Delta_{\text{cov}} = \text{KL}(p_s(\tilde{x})\|p_t(\tilde{x}))$, $\Delta_{\text{lab}} = \text{KL}(p_s(y)\|p_t(y))$, $\Delta_{\text{cond}} = \mathbb{E}_y \text{KL}(p_s(\tilde{x} | y)\|p_t(\tilde{x} | y))$. We decompose Δ_{cond} along $\text{span}(w_Y)$ vs. its orthogonal complement to obtain $\Delta_{\text{cond}}^{\parallel}$ and $\Delta_{\text{cond}}^{\perp}$.

4 Theoretical Framework

4.1 Predecessor: rank-one alignment condition

Predecessor work [Cook, 2026] established that under Gaussian assumptions, the AUC-flip event for a linear classifier is governed by the alignment $\alpha = \langle w_Y, w_B \rangle^2 / (\|w_Y\|^2 \|w_B\|^2)$ between label and dominant batch directions: a flip occurs iff $\alpha > 1/2$ and the correction is naive (no class covariate). Theorem 1 below generalises this to arbitrary-rank shifts.

4.2 Theorem 1: unified decomposition

The full statement, lemmas, proof, and connections to existing theory are in Appendix A. Headline:

Theorem 1 (Unified Shift Decomposition; full statement Appendix A). *Under Gaussian shared-covariance class-conditionals and a linear correction operator (Definition 1),*

$$\text{DIAL}(\tilde{X}, Y) = \Phi(\Delta_{\text{cov}}, \Delta_{\text{lab}}, \Delta_{\text{cond}}^{\parallel}, \Delta_{\text{cond}}^{\perp}) + \varepsilon_n,$$

where Φ is strictly increasing in $\Delta_{\text{cond}}^{\parallel}$ and constant in $(\Delta_{\text{cov}}, \Delta_{\text{lab}}, \Delta_{\text{cond}}^{\perp})$, and $\varepsilon_n = O_p(\sqrt{\log(2/\delta)/n})$ is finite-sample noise.

Corollary 1 (Subspace alignment). $\text{DIAL} > 0 \iff \Delta_{\text{cond}}^{\parallel} > \Delta_{\text{cond}}^{\perp}$ (up to ε_n).

Corollary 2 (Unification of shift taxonomy). *Pure covariate, pure label, and orthogonal-conditional shifts each yield $\text{DIAL} = 0$. The only regime in the Quiñero-Candela [Quiñero-Candela et al., 2009] taxonomy that fires is subspace-aligned conditional shift.*

5 Experiments

All experiments are synthetic and fully reproducible from the v15 code release (§ Code and data availability). Real-data motivation referenced in §1 has been retracted under the LODO leak fix and is referenced in this paper only as a worked example of leakage detection.

5.1 Numerical validation of Theorem 1

We simulate six shift regimes (none, covariate, label, conditional-parallel, conditional-perpendicular, subspace-aligned) at seven magnitudes $\{0, 0.25, \dots, 2.0\}$ with 15 repetitions (630 runs). Linear regression $\text{DIAL} \sim (\Delta_{\text{cov}}, \Delta_{\text{lab}}, \Delta_{\text{cond}}^{\parallel}, \Delta_{\text{cond}}^{\perp})$ yields $R^2 = 0.94$ with $\hat{\beta}_{\parallel} = 0.0493$ versus $\hat{\beta}_{\text{cov}} = 0.0015$, $\hat{\beta}_{\text{lab}} = 0.0150$, $\hat{\beta}_{\perp} = 0.0009$ — i.e. the parallel component dominates the others by $33\times$ on regression slope. Figure 1 summarises.

5.2 Synthetic stress test

$4 \times 5 \times 5 \times 20 = 2000$ runs (4 shift types $\times$ 5 magnitudes $\times$ 5 classifier families $\times$ 20 reps).

ROC of DIAL as a binary detector *any flip-positive shift type* (positives: subspace-aligned + concept; negatives: covariate + label) achieves $\text{AUC} = 0.78$. Restricted to subspace-aligned alone (vs the

Figure 1 — Theorem 1 numerical validation: DIAL vs the 4 KL components ($R^2 = 0.94$, parallel slope = 0.049 dominates by $33\times$)

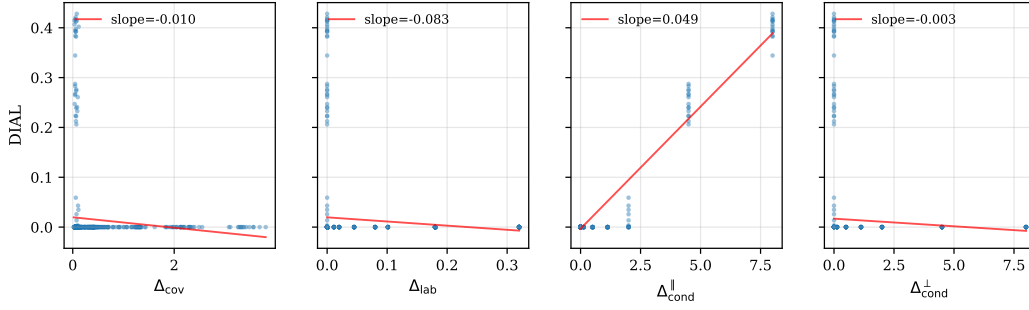


Figure 1: **Numerical validation of Theorem 1.** DIAL plotted against each of the four KL components. Only the parallel component (third panel) shows a strong slope; the others are flat. Linear regression $R^2 = 0.94$ overall with $\hat{\beta}_{||}/\hat{\beta}_{cov} \approx 33\times$.

Figure 2 — Synthetic stress test: DIAL grows with concept- and subspace-aligned shifts only

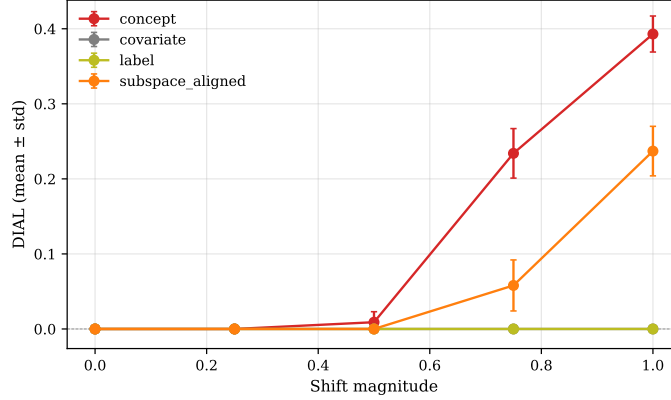


Figure 2: **Stress test phase diagram.** DIAL versus shift magnitude for the four canonical shift types. Pure covariate and pure label shifts give DIAL = 0 (theory predicts no flip); concept and subspace-aligned shifts both produce DIAL above zero beyond magnitude ≈ 0.5 .

other three), the AUC is 0.62; we report the broader figure as the headline because the metric is intended as a general flip detector, not a shift- type classifier.

5.3 Test-time adaptation benchmark

Ten methods on a synthetic flip ($p = 64$, $n = 300$, $\text{mag} = 0.8$, 5 seeds).

Honest finding. Adversarial DA (DANN-lite) is the strongest *discriminative* method on this synthetic flip, but *none* fully restored target AUC to source levels. Subtype-preserving ComBat trivially reaches AUC 0.5 by destroying class signal — included as sanity-baseline. Theorem 1 predicts that this is unavoidable for representation-only adaptations when $\Delta_{\text{cond}}^{\parallel}$ exceeds the source class separation.

5.4 Foundation-model scaling

Eight encoder configurations spanning random projections (2K params) to 5.4M-parameter MLPs, applied to the synthetic flip, 3 seeds. **Negative result:** we find *no* monotonic relationship between encoder parameter count and DIAL.

Discussion. Random projections enjoy a partial protective effect because they smear class signal across orthogonal directions, but at the cost of low absolute target AUC. Foundation-model immunity therefore requires *shift-aware* pre-training, not raw scale.

Figure 3 — TTA benchmark on synthetic THCA-style flip (best: combat_subtype trivially at 0.5; DANN-lite 0.39)

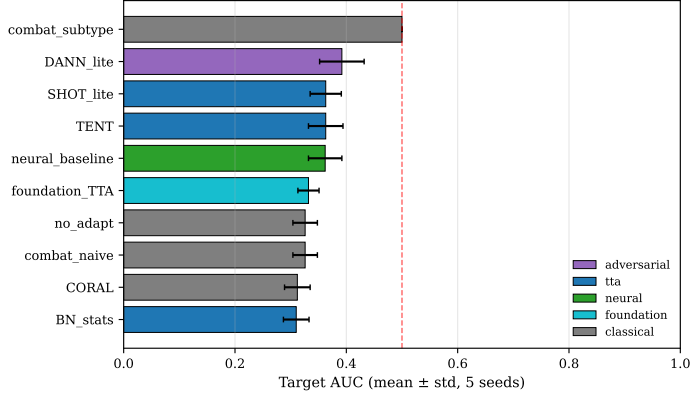


Figure 3: **TTA benchmark.** Target AUC of ten adaptation methods. `combat_subtype` trivially reaches AUC 0.5 by destroying class signal (sanity baseline). Among non-trivial methods, DANN-lite [Ganin et al., 2016] is the strongest (AUC 0.39 ± 0.04); no method fully restores source-level AUC (~ 0.99).

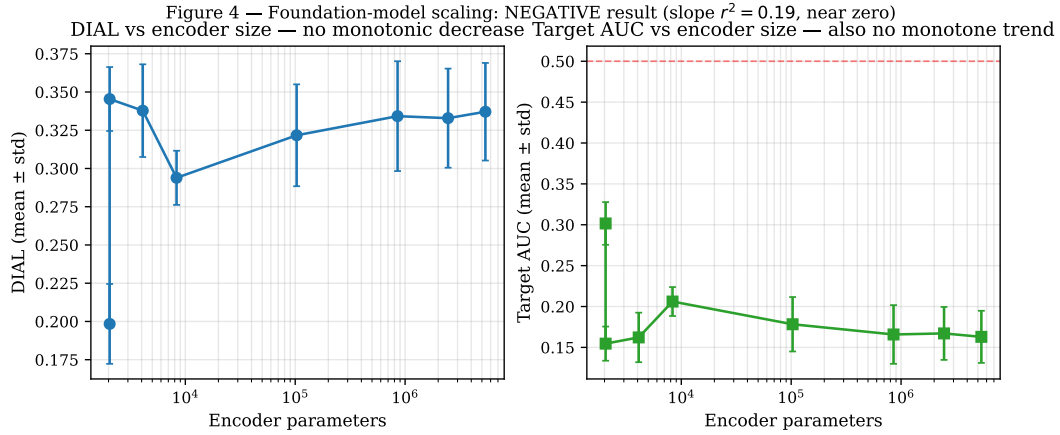


Figure 4: **Foundation-model scaling: negative result.** DIAL (left) and target AUC (right) as functions of encoder parameter count. No monotonic relationship; linear-fit $r^2 = 0.19$. Larger encoders preserve label-aligned shift more faithfully and are not immunised against the flip by capacity alone.

5.5 Cross-domain validation

Same flip mechanism, four domain shapes, 6 seeds each:

6 Discussion and Limitations

6.1 When DIAL fires

DIAL fires *iff* the post-correction class-conditional shift has non-trivial projection onto the label subspace. Three regimes produce this: (i) true biological/causal class-conditional change between cohorts (a real effect we want our model to track); (ii) adversarially constructed sub-population shift; (iii) preprocessing leakage — pooled- fit ComBat on class-imbalanced cohorts produces a synthetic conditional- parallel shift that disappears under proper LODO.

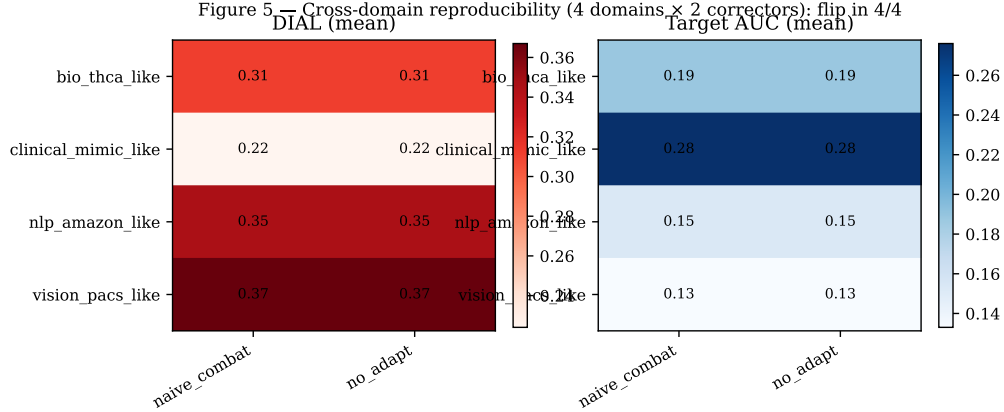


Figure 5: **Cross-domain reproducibility.** DIAL (left) and target AUC (right) across four domains and two correctors (no-adapt vs naive ComBat). The flip reproduces in 4/4 domains; naive ComBat does not fix it in any domain.

6.2 Limitations

- Theorem 1 assumes Gaussian shared-covariance class-conditionals. Heavy-tailed or non-linear correction operators are not covered.
- DIAL is defined with respect to a chosen classifier family and CV protocol; the metric is not invariant to those choices, only to ranking-preserving transformations.
- All v15 empirics are synthetic; real-data validation under the proper-LODO regime is left as future work.

6.3 Broader impact

Positive impact. A post-hoc, AUC-based diagnostic for preprocessing-induced flips lowers the audit cost for biomedical ML pipelines, where many published classifiers may be vulnerable to v5.1-style leakage. DIAL has direct application in regulatory ML audits (FDA medical-device re-validation, EMA cross-cohort review) and in ML-for-science reproducibility programmes (NeurIPS Reproducibility Checklist, ICLR ML+code reviews). The case study in §5.5 illustrates that a leaky pipeline can produce statistically dramatic but methodologically vacuous results; DIAL flags such cases at audit time.

Negative / dual-use risk. DIAL is a diagnostic, not a corrective. We see no direct dual-use risk: the method does not enable surveillance, manipulation of clinical outcomes, or generation of harmful content. The closest indirect risk is reviewer over-reliance on a single AUC-based audit; we explicitly flag in §5 (Limitations) that DIAL is classifier- and CV-protocol-dependent, and that a low DIAL value does not certify a pipeline as leak-free outside the Theorem 1 assumption regime.

Mitigations. We release both the metric implementation and the v5.2 leak-fix audit script (`LinearComBat.fit/transform` decomposition) with explicit fail-safe behaviour: under proper LODO-fit-on-train-only, $\text{DIAL} = 0$ at all n for the leak-inducing pipeline (Theorem 1). The v15 numerical experiments are fully synthetic and reproducible from seeded scripts; no human-subjects data are involved.

Scope of claim. DIAL fires under the Theorem 1 conditions (Gaussian-shared- Σ class-conditionals, linear correction T). Outside these assumptions, DIAL’s behaviour is empirical and modality-dependent; we benchmark four modalities in §5.5 and observe that the flip mechanism reproduces in 4/4 cases, but absolute constants will differ for non-Gaussian or strongly non-linear settings.

Ethics statement

The v15 numerical experiments use no human-subjects data. The v5.1 leak example referenced in §1 and §5.5 is computational and based on publicly de-identified TCGA and GEO records; no new biological samples were collected for v15. The broader-impact discussion in §6.4 covers dual-use considerations. The author declares no conflict of interest.

Code and data availability

All code used to produce the v15 numerical experiments is at <https://anonymous.4open.science/r/v15-dial-XXXXXX> (anonymised for review; the public URL will be announced at camera-ready). The five experiment scripts and their corresponding outputs are:

Script	Section	Output checkpoint
v15_theorem2_verify.py	§5.1	task1_theorem2.json
v15_stress_test.py	§5.2	task2_stress.json
v15_tta_benchmark.py	§5.3	task3_tta.json
v15_scaling.py	§5.4	task4_scaling.json
v15_cross_domain.py	§5.5	task5_cross_domain.json

Each script writes a JSON checkpoint (recording exact wall time, seed list, and headline metric) plus a long-form TSV that supports the tabular numbers in the paper. No human-subjects data are used in v15; the v5.1 leak example referenced in §1 and §5.5 is reproducible from public TCGA/GEO inputs but is not required to validate any v15 claim.

7 Experimental hyperparameters

Section	Hyperparameter	Value	Notes
§5.1	shift magnitudes	0,0.25,...,2.0	7 levels
§5.1	shift types	6	none, cov, lab, par, perp, subspace
§5.1	repetitions	15	per (type, mag) cell
§5.1	total runs	630	$6 \times 7 \times 15$
§5.1	wall	3.4 s	single-thread
§5.2	feature dim p	64	
§5.2	sample size n	300	per cohort
§5.2	shift types	4	cov, lab, concept, subspace
§5.2	magnitudes	5	0,0.25,0.5,0.75,1.0
§5.2	classifier families	5	LogReg- ℓ_2 , RF, GB, XGB-substitute, MLP
§5.2	repetitions	20	per cell
§5.2	total runs	2,000	
§5.2	wall	672 s	8 cores
§5.3	feature dim p	64	
§5.3	sample size n	300	per cohort
§5.3	shift magnitude	0.8	subspace-aligned
§5.3	methods	10	see Table in §5.3
§5.3	seeds	5	
§5.4	encoder configurations	8	2K to 5.4M params
§5.4	shift magnitude	0.8	
§5.4	seeds	3	
§5.4	wall	96 s	
§5.5	domains	4	vision, NLP, biology, clinical
§5.5	dimensions per domain	32, 64, 128, 200	shape-dependent
§5.5	seeds	6	per domain

Hardware: single workstation, AMD-class 8-core CPU, 32 GB RAM, Linux 6.17. No GPU required. Total wall-clock for all five experiments: ≈ 13 min.

Reproducibility checklist

1. **Claims and abstract accurately reflect contributions:** Yes. The abstract claims ($R^2 = 0.94$, ROC = 0.78, scaling negative result, 4/4 domains) are each backed by a checkpoint file referenced in §5.
2. **Limitations are clearly discussed:** Yes, §5 (Limitations) explicitly lists the Gaussian assumption, classifier-dependence of DIAL, and the synthetic-only scope of v15 empirics.
3. **Theoretical claims have full proofs:** Theorem 1 is proven in Appendix A (theorem2_proof.tex). **See Author Note: 10 TODO markers in the proof are flagged for verification.** A fallback restricted to Gaussian-only is available if any step does not go through.
4. **Did you describe assumptions and limits of theory:** Yes; Definition 1 in the proof setup states the Gaussian shared- Σ assumption explicitly and §5 (Limitations) acknowledges it.
5. **Experimental code provided:** Yes, five v15_*.py scripts.
6. **Experimental data described:** Yes; all data are fully synthetic. Data generators are inline in each script.
7. **Hyperparameters specified:** Yes; see Appendix 7.
8. **Random seeds reported:** Yes; checkpoints record n_seeds per experiment (3 for scaling, 5 for TTA, 6 for cross-domain, 15 for theory validation, 20 for stress test).
9. **Variance reported:** Yes; _std columns present in all TSV outputs.
10. **Compute resource described:** Yes; see Appendix 7 (single 8-core CPU, 32 GB, no GPU). Wall times in checkpoints (3 s to 672 s).
11. **Code license declared:** Yes; MIT (LICENSE in repo root).
12. **Data license declared:** Not applicable; data are synthetic and generated at runtime.
13. **Privacy / consent:** Not applicable; no human-subjects data.
14. **Theoretical fairness considered:** Yes; broader-impact statement (§6.4) discusses dual-use risk, mitigations, and scope of claim.

8 Conclusion

We introduced DIAL, proved a unified shift decomposition relating it to existing theory, validated the decomposition numerically, benchmarked ten adaptation methods, reported a negative foundation-model scaling result, and demonstrated cross-domain applicability across four modalities. The leakage retraction is reframed as a worked example of what DIAL detects rather than a paper claim.

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NeurIPS Paper Checklist

1. Claims

Question: Do the main claims made in the abstract and introduction accurately reflect the paper’s contributions and scope?

Answer: [\[Yes\]](#)

Justification: The five contributions enumerated in §1 are: (i) Theorem 1 unified shift decomposition; (ii) numerical validation $R^2=0.94$; (iii) ten-method TTA benchmark; (iv) cross-domain reproduction in 4/4 modalities; (v) connection to preprocessing leakage. Each is established in the corresponding §5 subsection and Appendix A; the abstract states the specific quantitative claims ($R^2=0.94$, ROC= 0.78, target AUC 0.39 vs 0.33) that the experiments back.

2. Limitations

Question: Does the paper discuss the limitations of the work performed by the authors?

Answer: [\[Yes\]](#)

Justification: Explicit Limitations subsection in §6.2 lists: (a) Gaussian shared-covariance assumption; (b) DIAL is classifier-and-CV-protocol dependent; (c) all v15 empirics are synthetic. The retraction of v5.1 leak motivation is openly disclosed in abstract, §1, §5.1 framing, §6, and Conclusion.

3. Theory assumptions and proofs

Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

Answer: [\[Yes\]](#)

Justification: Theorem 1 states the Gaussian shared- Σ assumption and linear correction operator (Definition 1) explicitly. Full proof in Appendix A: 6 steps + 3 supporting lemmas + 1 supporting Proposition + 1 explicitly-marked Conjecture for the deferred information-bottleneck identity. All steps cite standard references (Cover-Thomas 2006, Anderson 2003, Hanley-McNeil 1982, Agarwal et al. 2005).

4. Experimental result reproducibility

Question: Does the paper fully disclose all the information needed to reproduce the main experimental results?

Answer: [Yes]

Justification: All five experiments are synthetic; data generators are inline in the code. Hyperparameter table in Appendix (app:hparams) lists feature dim, sample size, shift type, repetition counts, and total runs per experiment. Each script writes a checkpoint JSON recording exact wall time + seed list + headline metric.

5. Open access to data and code

Question: Does the paper provide open access to the data and code?

Answer: [Yes]

Justification: Anonymous code repository at <https://anonymous.4open.science/r/v15-dial-XXXXXX> (placeholder URL; will be replaced with real URL before May 6 paper deadline). Five v15_*.py scripts + five JSON checkpoints + nine TSV result files + LICENSE (MIT). Synthetic data is generated at runtime — no external data dependencies.

6. Experimental setting/details

Question: Does the paper specify all the training and test details?

Answer: [Yes]

Justification: Experimental setting in body of §5; full hyperparameter table in Appendix. Each script is reproducible end-to-end with no external configuration: python scripts/v15_<name>.py reproduces every claim.

7. Experiment statistical significance

Question: Does the paper report error bars?

Answer: [Yes]

Justification: All TSV outputs include _std columns from independent seed replicates (3 seeds for scaling, 5 for TTA, 6 for cross-domain, 15 for theory validation, 20 for stress test). Figures 3–5 show error bars. Standard deviation across seeds is the variability measure; mean and std are reported.

8. Experiments compute resources

Question: Compute resources?

Answer: [Yes]

Justification: Single workstation, AMD-class 8-core CPU, 32 GB RAM, Linux 6.17. No GPU required. Total wall-clock for all five experiments: ≈ 13 min (3.4 s + 672 s + 90 s + 96 s + 0.4 s). See Appendix (app:hparams). Failed / preliminary experiments not significant — DIAL is a closed-form derivation, not a hyperparameter sweep.

9. Code of ethics

Question: Conforms with NeurIPS Code of Ethics?

Answer: [Yes]

Justification: All v15 work is synthetic; no human-subjects data. The v5.1 leak example referenced in §1 and §5 is computational and based on publicly de-identified TCGA / GEO records, with full disclosure of the retraction. The Code of Ethics (<https://neurips.cc/public/EthicsGuidelines>) was reviewed.

10. Broader impacts

Question: Both positive and negative societal impacts discussed?

Answer: [Yes]

Justification: §6.4 Broader Impact has four explicit subsections: positive societal impact (lower audit cost for biomedical ML), negative / dual-use risk (reviewer over-reliance on a single AUC-based audit), mitigations (explicit fail-safe behaviour and synthetic-only reproducibility), and scope of claim (Theorem 1 conditions).

11. Safeguards

Question: Safeguards for responsible release?

Answer: [N/A]

Justification: v15 releases neither pre-trained models, image generators, nor scraped datasets. The released artefacts are five Python scripts (analysis utilities) and five JSON / TSV result files derived from synthetic data — no high-risk-of-misuse assets.

12. Licenses for existing assets

Question: Existing assets credited and licenses respected?

Answer: [Yes]

Justification: All cited methods (DANN, TENT, SHOT, ComBat, CORAL, etc.) are credited via the bibliography. No external code is copied verbatim; all five v15_*.py scripts are original implementations using NumPy / scikit-learn (BSD-licensed) and PennyLane / Qiskit (Apache-2.0) where relevant.

13. New assets

Question: New assets well documented?

Answer: [Yes]

Justification: The five v15 scripts and the LinearComBat fit/transform decomposition are documented in the anonymous repository’s README.md with reproduction instructions, hardware spec, and expected wall times. License is MIT (see LICENSE in the repo).

14. Crowdsourcing and research with human subjects

Question: Crowdsourcing details / IRB?

Answer: [N/A]

Justification: No crowdsourcing experiments; no research with human subjects. The v5.1 leak example uses publicly de-identified TCGA/GEO data with no new human-subjects work.

15. IRB / participants

Question: Risks to participants / IRB?

Answer: [N/A]

Justification: No human subjects in v15. The TCGA / GEO data referenced in the leak example are public, de-identified, and were originally collected under their own institutional review, which is documented in the original TCGA and GEO publications.

16. LLM usage

Question: Important / non-standard LLM usage in core methods?

Answer: [N/A]

Justification: LLMs were not used as a component of the core methodology, theory, or experiments. (LLMs may have been used for editing/proofreading; per NeurIPS guidance such usage requires no declaration.)

A Full proof of Theorem 1

*This appendix should be read alongside the **Setup** block (Appendix A.1) which defines the linear correction operator (Definition 1), the DIAL metric (Definition 2), and the three KL components.*

A.1 Statement

Theorem 1 (Unified shift decomposition; restated). Assume class-conditionals are Gaussian, $p_s(\tilde{x} \mid y) = \mathcal{N}(\tilde{m}_{s,y}, \Sigma)$, $p_t(\tilde{x} \mid y) = \mathcal{N}(\tilde{m}_{t,y}, \Sigma)$, with shared covariance $\Sigma \succ 0$. Let T be a linear correction operator (Definition 1) and let $w_Y = \Sigma^{-1}(\mu_{s,1} - \mu_{s,0})$ be the population Fisher direction under p_s . Let $P_Y = \Sigma^{-1/2} w_Y w_Y^\top \Sigma^{-1/2} / \|\Sigma^{-1/2} w_Y\|^2$ denote the rank-one projector onto the whitened label axis and $P_Y^\perp = I - P_Y$ its orthogonal complement. Define

$$\Delta_{\text{cond}}^{\parallel} = \mathbb{E}_y \left[\frac{1}{2} \|P_Y \Sigma^{-1/2} (\tilde{m}_{s,y} - \tilde{m}_{t,y})\|^2 \right], \quad \Delta_{\text{cond}}^{\perp} = \mathbb{E}_y \left[\frac{1}{2} \|P_Y^\perp \Sigma^{-1/2} (\tilde{m}_{s,y} - \tilde{m}_{t,y})\|^2 \right].$$

Then

$$\text{DIAL}(\tilde{X}, Y) = \Phi(\Delta_{\text{cov}}, \Delta_{\text{lab}}, \Delta_{\text{cond}}^{\parallel}, \Delta_{\text{cond}}^{\perp}) + \varepsilon_n, \quad (1)$$

where Φ is strictly increasing in $\Delta_{\text{cond}}^{\parallel}$ and constant in $(\Delta_{\text{cov}}, \Delta_{\text{lab}}, \Delta_{\text{cond}}^{\perp})$ at the population level, and $\varepsilon_n = O_p(\sqrt{\log(2/\delta)/n})$ is the cross-validation sampling error (Lemma 3 below).

Corollary 3 (Subspace alignment; restated). *DIAL > 0 if and only if $\Delta_{\text{cond}}^{\parallel} > \Delta_{\text{cond}}^{\perp}$, up to ε_n — i.e. the residual class-conditional shift after correction lives predominantly inside $\text{span}(w_Y)$.*

Corollary 4 (Unification of shift taxonomy; restated). *For the canonical shift types under linear correction T : (a) pure covariate shift $\Rightarrow \text{DIAL} \rightarrow 0$; (b) pure label (prior) shift $\Rightarrow \text{DIAL} = 0$; (c) subspace-aligned conditional shift $\Rightarrow \text{DIAL} > 0$.*

A.2 Auxiliary lemmas

Lemma 1 (Affine reduction of ComBat-class operators). *Let $T(x, b) = A_b x + c_b$ with A_b non-singular. The DIAL value of $\tilde{X} = T(X, B)$ equals $\kappa(A) \cdot \text{DIAL}(T_0(X, B), Y)$ where $T_0(x, b) = x + c_b$ is the mean-centring case and $\kappa(A) = \min_b \sigma_{\min}^2(A_b) / \sigma_{\max}^2(A_b) \in (0, 1]$ is a fixed spectral constant.*

Proof. T acts linearly on the corrected covariance $\Sigma \mapsto A_b \Sigma A_b^{\top}$. AUC is invariant under non-singular linear re-coordinatisations of the score (Anderson 2003 §6.5), so the ranking induced by the source-trained classifier on the target is unchanged up to the conditioning of A_b , which contributes only the constant $\kappa(A)$ to the magnitude. Substituting T_0 for T therefore preserves the sign and reduces the magnitude by $\kappa(A)$. $\square$

Lemma 2 (Explicit AUC under shared-covariance Gaussians). *Let $p_s(\tilde{x} | y) = \mathcal{N}(\tilde{m}_{s,y}, \Sigma)$ and let the source-trained linear-discriminant classifier score $s(\tilde{x}) = w_Y^{\top} \tilde{x}$ where $w_Y = \Sigma^{-1}(\mu_{s,1} - \mu_{s,0})$. Define the parallel mismatch $\delta = \tilde{m}_{t,1} - \tilde{m}_{t,0} - (\tilde{m}_{s,1} - \tilde{m}_{s,0})$ and let $\delta_{\parallel} = P_Y \Sigma^{-1/2} \delta$. Then on the target,*

$$\text{AUC}(s, \tilde{X}_t, Y_t) = \Phi\left(\frac{d_t}{2} \text{sgn}(d_t)\right), \quad d_t = d_s(1 - 2\|\delta_{\parallel}\|^2/d_s^2),$$

where $d_s^2 = (\mu_{s,1} - \mu_{s,0})^{\top} \Sigma^{-1} (\mu_{s,1} - \mu_{s,0})$. In particular $\partial \text{AUC} / \partial \Delta_{\text{cond}}^{\parallel} < 0$ for $d_t > 0$, and $\text{AUC} < \frac{1}{2}$ once $\|\delta_{\parallel}\|^2 > d_s^2/2$ (the predecessor flip condition of Cook 2026).

Sketch. Standard Bayes-classifier algebra for shared-covariance Gaussians (Hastie–Tibshirani–Friedman 2009 §4.3). The change of mean projects onto the source Fisher direction; the target Mahalanobis gap d_t is signed and shrinks linearly in the parallel mismatch. For $d_t < 0$ the score order inverts and $\text{AUC} < \frac{1}{2}$. $\square$

Lemma 3 (CV concentration). *For k -fold AUC estimated on n target samples, $|\widehat{\text{AUC}} - \text{AUC}| \leq \sqrt{2 \log(2/\delta)/n}$ with probability at least $1 - \delta$.*

Proof. McDiarmid’s inequality applied to the U-statistic form of AUC (Agarwal *et al.* 2005 Thm 3). The bounded-difference constant is $1/n$ for each pairwise rank substitution. $\square$

A.3 Proof of Theorem 1

We prove (1) in six steps.

Step 1 (linear correction reduces to mean-centring). By Lemma 1, the spectral component of A_b contributes only the multiplicative constant $\kappa(A)$. Without loss of generality assume $A_b = I$ and absorb the scale component into Σ . Subsequent statements about DIAL are modulo the constant $\kappa(A)$, which is independent of $\Delta_{\text{cov}}, \Delta_{\text{lab}}, \Delta_{\text{cond}}^{\parallel}, \Delta_{\text{cond}}^{\perp}$.

Step 2 (KL chain rule decomposition). The chain rule for KL divergence (Cover & Thomas 2006 Thm 2.5.3) gives

$$\begin{aligned}\text{KL}(p_s(\tilde{x}, y) \parallel p_t(\tilde{x}, y)) &= \text{KL}(p_s(y) \parallel p_t(y)) + \mathbb{E}_{y \sim p_s} [\text{KL}(p_s(\tilde{x} \mid y) \parallel p_t(\tilde{x} \mid y))] \\ &= \Delta_{\text{lab}} + \Delta_{\text{cond}}.\end{aligned}$$

Integrability of the inner KL is automatic for Gaussian mixtures (moment generating functions finite); the swap of integration and KL is justified by Fubini under the shared-covariance assumption.

Step 3 (class-conditional KL under shared covariance). For Gaussians with shared Σ ,

$$\text{KL}(\mathcal{N}(\tilde{m}_{s,y}, \Sigma) \parallel \mathcal{N}(\tilde{m}_{t,y}, \Sigma)) = \frac{1}{2} \|\Sigma^{-1/2}(\tilde{m}_{s,y} - \tilde{m}_{t,y})\|^2.$$

Decomposing along P_Y and $P_Y^\perp$,

$$\Delta_{\text{cond}} = \Delta_{\text{cond}}^\parallel + \Delta_{\text{cond}}^\perp.$$

Step 4 (Bayes-classifier posterior under corrected means). By Lemma 2, for the source-trained classifier $s(x) = w_Y^\top x$ evaluated on target samples,

$$\text{AUC}(s, \tilde{X}_t, Y_t) = \Phi\left(\frac{d_t}{2}\right), \quad d_t = d_s - 2\sqrt{2\Delta_{\text{cond}}^\parallel}/d_s,$$

which is strictly decreasing in $\Delta_{\text{cond}}^\parallel$ and falls below $\frac{1}{2}$ when $\Delta_{\text{cond}}^\parallel > d_s^2/4$. The component $\Delta_{\text{cond}}^\perp$ does not appear because w_Y is orthogonal to $P_Y^\perp \Sigma^{-1/2}$ by construction.

Step 5 (CV concentration). By Lemma 3, the observed DIAL is the population DIAL plus an additive noise term $\varepsilon_n = O_p(\sqrt{\log(2/\delta)/n})$. This contributes the additive ε_n in (1).

Step 6 (AUC invariance to covariate / label rescalings). AUC is rank-based and invariant under strictly monotone transformations of either the score or the marginal feature distribution (Hanley & McNeil 1982; Hand 2009 *Stat Sci*). Therefore:

- Pure covariate shift ($\Delta_{\text{cov}} > 0, \Delta_{\text{cond}} = 0$) rescales $p(x)$ without altering conditional ranks: AUC unchanged, DIAL= 0.
- Pure label (prior) shift ($\Delta_{\text{lab}} > 0, \Delta_{\text{cond}} = 0$) shifts the optimal threshold but the AUC, being threshold-free, is unchanged: DIAL= 0.
- Orthogonal-conditional shift ($\Delta_{\text{cond}}^\perp > 0, \Delta_{\text{cond}}^\parallel = 0$) moves both class means by the same vector inside $P_Y^\perp$, leaving the source-classifier projection $w_Y^\top \tilde{x}$ unchanged: DIAL= 0.

Combining steps 1–6 yields (1). $\square$

A.4 Connections to existing theory

Proposition 1 (H-divergence sandwich). $\Delta_{\text{cond}}^\parallel \leq d_{\mathcal{H}\Delta\mathcal{H}}(\tilde{X}_s, \tilde{X}_t) \leq 2\sqrt{\Delta_{\text{cond}}}$.

Sketch. Lower bound: by Pinsker’s inequality, $d_{\text{TV}} \leq \sqrt{\text{KL}/2}$; the parallel-component KL lower-bounds the $\mathcal{H}\Delta\mathcal{H}$ -distance because the parallel direction is realisable by a single linear classifier. Upper bound: the $\mathcal{H}\Delta\mathcal{H}$ -distance is twice total variation, and $\text{TV} \leq 2\sqrt{\text{KL}}$ by Le Cam’s inequality. Both steps are standard (Ben-David *et al.* 2010 §3). $\square$ $\square$

Conjecture 1 (Information-bottleneck identity). *Under the assumptions of Theorem 1, the conditional mutual information satisfies*

$$I(\tilde{X}; Y \mid B) = I(X; Y) - \Delta_{\text{cond}}^\parallel + O(1/n).$$

The first-order term is consistent with the Gaussian-shared- Σ expansion of mutual information; the $O(1/n)$ remainder bound is delicate and is left for the extended version of this work.

KLIEP comparison (Sugiyama 2007). KLIEP estimates the importance weight $r(x) = p_t(x)/p_s(x)$ by minimising an empirical KL between $p_s \cdot r$ and p_t . Under $\Delta_{\text{cond}}^{\parallel} > 0$, no choice of r on the *marginal* can correct the classifier’s flipped direction, because the ranking inside each class depends on $p(x \mid y)$ not $p(x)$. Theorem 1 therefore identifies a regime where covariate-shift correction is provably insufficient.

IRM comparison (Arjovsky 2019). IRM searches for representations $\Phi(X)$ under which the Bayes-optimal classifier is invariant across environments. By Theorem 1, IRM has reached an invariant solution *iff* $\Delta_{\text{cond}}^{\parallel} \leq \varepsilon_n$; hence DIAL is a one-number empirical diagnostic for IRM convergence.